



Jargon Buster

Anti-aliasing: Anti-aliasing is a technique used to reduce the visual impact of aliasing, or jagged edges, on a computer graphics system.

Anisotropic filtering: An advanced texture-filtering technique that improves image quality of objects that extend from the foreground to deep into the background.

Fill rate: The speed at which a graphics card can render pixels, usually measured in Megapixels per second. GPUs with higher fill rates can display higher resolutions and more colours, at higher frame rates than other chips with lower fill rates.

Frame buffer: The memory that is dedicated to the graphics processor and used to store rendered pixels before they're displayed on the monitor.

Frames per second (fps): The rate at which the graphics processor renders new frames, or screens full of pixels. A

faster GPU renders more frames per second, making the application more fluid and responsive to user input.

Graphics Processing Unit (GPU): A high-performance 3D processor that integrates the entire 3D pipeline—transformation, lighting, setup and rendering. A GPU offloads all 3D calculations from the CPU, freeing the CPU for other functions such as physics and artificial intelligence.

Lighting: In 3D graphics, lighting is used to calculate the degree to which an object is affected by a source, and is used in games to create realistic-looking scenes with greater depth, instead of flat-looking or cartoonish environments.

Pixel: Stands for 'picture element'. A pixel is the smallest element of a graphics display.

Pixel shaders: Pixel shaders alter lighting and surface effects, which gives a natural look to materials and surfaces.

Polygon: The building block of all 3D objects, usually triangles or rectangles, used to form the surfaces and skeletons of 3D objects.

Texel: Texels are the smallest units of a texture map, similar to pixels.

Texture: An image file that is used to add complex patterns to the surfaces of objects in a 3D scene.

Texture mapping: The process of applying a texture to the surface of 3D models in order to simulate walls, skies, etc. Texture mapping enables developers to add realism to their models.

Vertex shaders: Vertex Shaders are used to give a life-like look to characters and environments. For example, developers can add wrinkles or dimples using vertex shaders—when a character smiles.

GDDR3: A newer memory standard that has been co-developed by ATI and JEDEC, and is an open standard now. It yields greater performance with lower power consumption.

9.0. Since the GeForce 6800 Ultra does, this opens up the possibilities of more dynamic programming control.

Memory counts

After all the hype about increased computational performance, there's an even more important factor—memory. Graphics memory has now moved on beyond DDR, and both ATi and nVidia have implemented the spanking new GDDR3 memory standard.

Even more pertinent is how ATi has worked with the memory. The Radeon 9800 XT's problem was the relatively slower memory clock—730 MHz, as compared to the GeForce FX 5950's 950 MHz. They finally set that right by implementing a brand new memory interface. It's a 256-bit-wide interface, similar to the Radeon 9800 XT. ATi claims that this inter-

face is more efficient, and that higher memory clock speeds are supported.

Thus far, the highest end X800 has 256 MB of GDDR3 that runs at 1,120 MHz,

yielding a peak memory bandwidth of 33.4 GBps. The interface has four 64-bit data channels communicating with four independent memory controllers, via a switch. ATi says that the memory interface can be extensively tuned; furthermore, the memory controllers can be reconfigured to perform more efficiently for certain applications.

The GeForce 6800 Ultra, too, uses GDDR3; the 256-bit wide memory bus

lem with the smaller 128-bit memory bus width; the FX 5950 changed that to 256-bit. But the biggest change with the GeForce 6800 Ultra is the potent combination of the wide 256-bit bus, the sixteen-pipeline design and the GDDR3 memory, which give it the large memory bandwidth.

It looks like GDDR3 is the way to go for the present and the future. The memory interface, type and the architectural design are going to affect the actual pixel and textured pixel (texel) fill rates, but with the kind of specs the Radeon X800 and the GeForce 6800 Ultra sport, there isn't going to be any bottlenecking there.

Performance optimisations

Neither ATi, nor nVidia have spared any other performance optimisations. ATi has introduced a new compression algorithm for normal maps, called 3Dc. The normal maps work on the idea that information about an object is stored as a texture, and each texture element stores three components of a vector perpendicular to the object surface at a given point. The DirectX 9.0 specification offers compression algorithms for this as well, but those would lead to information loss, with blocky artefacts in the stored normal maps. The X800 also features enhanced



The GeForce 6800 Ultra. Note the large heatsink and fan combination, and the two DVI connectors

clocks at 1,100 MHz, and the Ultra comes with 256 MB of memory onboard. The memory bandwidth is a good 35.2 GBps, as against the FX 5950's 28.3 GBps, which also has a 256-bit memory interface running 256 MB of DDR memory at 950 MHz.

Earlier examples of the NV3x architecture, such as the FX5800, had a prob-

Courtesy: FiringSquad.com

Info

The ATi Radeon 9800 XT was codenamed the R360, and the cutting edge Radeon X800 was codenamed the R420